

Full-Scope MCTS Output Review

Fix Status Report

This report covers the eight items raised in your review of the first full-scope MCTS output, plus one additional issue we found and fixed along the way. Eight of the nine are fixed and verified. The remaining item -VSA (partner) service schedule drift -is under active investigation; we are evaluating a few possible approaches and will share the details in a follow-up note shortly.

Summary

#	Item	Status
1	BBX2/BBX3 vessel capacity	Fixed
2 & 3	BMX rotation duration / wait & manoeuvring times	Fixed
4	VSA service schedule drift	In Progress
5	BBX speed exceeding the soft-cap rule	Fixed
6	Ports added with no cargo movement	Fixed
7	Cargo flow routing not visible in output	Fixed
8	Column definitions (vrank_mix, capacity)	Complete
9* (Additional)	FIX-type services not correctly locked	Fixed

Item 1 -BBX2/BBX3 vessel capacity

Status: Fixed

Observed

BBX2 was downgraded to VRank 2 (742 TEU nominal), but the 735 TEU reserved for external slotters wasn't reflected. BBX3's effective capacity (2,316 TEU) didn't reconcile with how BBX2 was calculated.

What we found

Capacity was being calculated with one flat number per vessel rank, without factoring in each line's own reserved/slotted capacity from the proforma data.

What we did

Each line's actual capacity discount and reserved-capacity figures are now factored into the capacity calculation. Verified against the source data -BBX2's 735 TEU reserved figure now matches exactly.

Item 2 & 3 -BMX rotation duration / wait & manoeuvring times

Status: Fixed

Observed

After JPYOK was added to BMX, wait and manoeuvring times showed as 0 (proforma states 197 hours), and the resulting rotation duration (25.77 days) wasn't a multiple of 7.

What we found

The wait/manoeuvring time figures were matched to each port by its position in the rotation list, not by which port it actually was. Once a port is added, the positions shift and stop lining up -the export fell back to zero for every port on the line, not just the new one. The 25.77-day figure was a manual sum of those zeroed columns; the underlying schedule constraint was never actually affected.

What we did

Ports are now matched to their timing data by port identity, not position. A newly-added port with no prior record now receives its own real measured timing data instead of a default of zero. Verified against JPYOK on BMX -its values now match the source data exactly, and every other port on the line keeps its original figures unchanged.

Item 4 -VSA service schedule drift

Status: In Progress

Observed

VSA (partner-operated) services are supposed to be frozen, yet their speeds, stay times, and sailing times appeared to be changing.

What we found

Rotation, weekly schedule, and vessel type are correctly locked for VSA lines. Sailing speed is intentionally left flexible, locking it conflicts with the partner's own schedule data and prevents the model from finding a valid solution at all. Port stay times, however, are not yet locked the same way, which is the likely source of the drift.

What we did

We are evaluating a few possible approaches to lock stay times without reintroducing the same conflict we saw when speed was locked outright. This is the one item still in progress -we will share the approach and next steps in a follow-up note in the next couple of days.

Item 5 -BBX speed exceeding the soft-cap rule

Status: Fixed

Observed

BBX was observed sailing at a constant 19 knots, appearing to exceed the 16.5 knot soft-cap speed rule.

What we found

The soft-cap penalty does exist and was applying correctly in principle, but a calibration error meant it started discouraging speed too early (13.5kt instead of 16.5kt) -the penalized range was roughly twice as wide as intended.

What we did

The calibration is corrected: the fuel penalty now starts exactly at the intended threshold. We are continuing to monitor speed behaviour as we test at larger network scale, as part of ongoing validation.

Item 6 -Ports added with no cargo movement

Status: Fixed

Observed

CNCWN was added to CP2 and MYLBU to YCX with no change in filling factor and zero ops time at either port -no apparent benefit to adding them.

What we found

The model checked that a candidate route was structurally valid, but never checked whether the added port would actually move any cargo once the network was solved.

What we did

A new check now runs after each candidate port addition is solved: if the port ends up moving essentially no cargo, that addition is rejected and not proposed again. Verified against real route data.

Item 7 -Cargo flow routing not visible in output

Status: Fixed

Observed

Cargo flow routing information was needed to properly review the quality of the output, particularly how demand was being routed.

What we found

The routing was already being calculated internally -it simply wasn't included in the exported output file.

What we did

A new cargo flow routing sheet has been added to the output workbook, showing routing by origin/destination pair and path. Verified that routes with real cargo flow populate correctly.

Item 8 -Column definitions (vrank_mix, capacity)

Status: Complete

Observed

Clarification was requested on what "vrank_mix" means, and the difference between "capacity" and "weekly_capacity_teu".

What we found

"vrank_mix" shows the mix of vessel types used on a line when more than one type is in service. Separately, "capacity" and "weekly_capacity_teu" were found to be showing identical values due to an export issue.

What we did

"vrank_mix" is documented for clarity (no change needed). The capacity/weekly_capacity_teu duplicate-value issue is fixed -"capacity" now correctly shows total fleet capacity for the line, distinct from the per-week figure.

Additionally found and fixed

While working through the items above, we also found and fixed one more issue you had not flagged, related to your item 4 above.

Item 9 -FIX-type services were not correctly locked

Status: Fixed

Observed

Not directly reported, but related to item 4 above.

What we found

Your service network has three types of lines: VSA (partner-operated, fully locked), FIX (operated by CNC but with a fixed, specialised port rotation), and OWN (fully flexible). The model was only recognising VSA and OWN -the 8 FIX-type lines were being treated as fully flexible, meaning their port rotations could be changed even though they should stay fixed.

What we did

FIX-type lines now have their port rotation locked, the same as VSA, while still allowing vessel size and count to be optimised on them, as intended. Verified against the source data for all 8 affected lines.